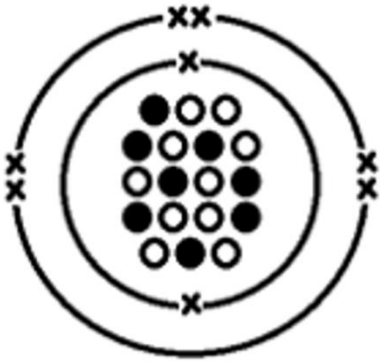
Chapter 5 – Atomic Structure

Atomic Structure and Atomic Symbol

Atomic Structure

Subatomic particle	Proton	Neutron	Electron
Relative mass	1	1	Negligible
Charge	$+1$	0	-1
Position	Nucleus	Nucleus	Moves around nucleus

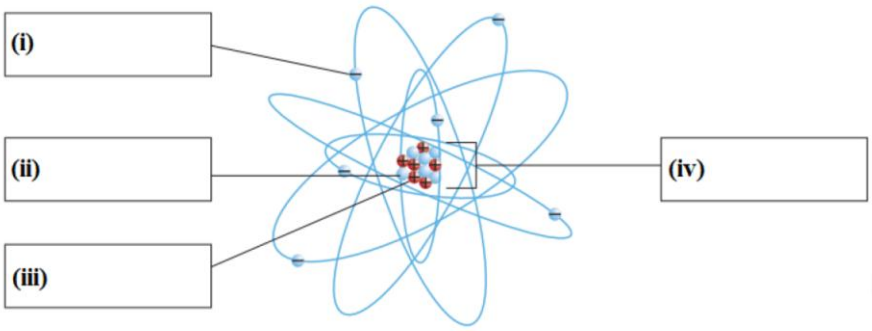


Definitions:

Element is a substance that is made up of only one kind of atoms.

Atom is the smallest part of an element which has the chemical properties of that element.

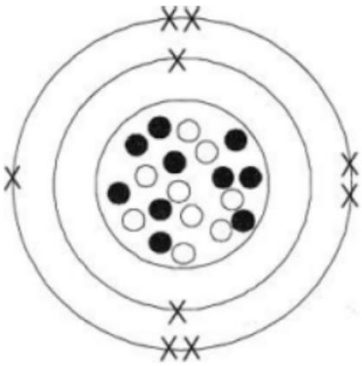
Structure Questions (Part1)



Q1: Why atom X is a electrically neutral?

Ans:_____

The following diagram shows the structure of an atom of element X.



Q2:

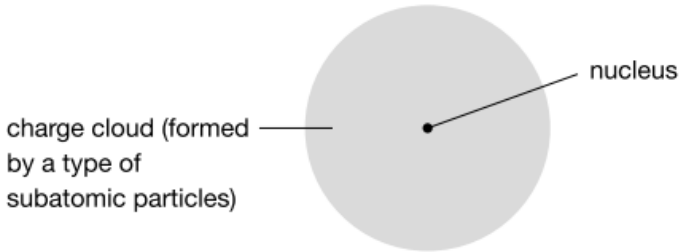
Identify the following sub-atomic particles

(i) Cross:

(ii) Black dot:

(iii) White dot:

Q3: The following simplified diagram represents the structure of the atom of a certain element with an atomic number greater than one.



(a) Does the atom carry any net charges? Suggest why.

(1 mark)

(b) What type of subatomic particles can be found in the charge cloud?

(1 mark)

(c) Name TWO types of subatomic particles that can be found in the nucleus.

(2 marks)

(d) Explain why most of the mass of the atom is in the nucleus.

(2 marks)

Atomic Symbol

Atomic number (No. of protons)	Atom	Atomic symbol	Electronic arrangement	Atomic number (No. of protons)	Atom	Atomic symbol	Electronic arrangement
1	Hydrogen*	H	1	11	Sodium#	Na	2,8,1
2	Helium*	He	2	12	Magnesium#	Mg	2,8,2
3	Lithium#	Li	2,1	13	Aluminium#	Al	2,8,3
4	Beryllium#	Be	2,2	14	Silicon^	Si	2,8,4
5	Boron^	B	2,3	15	Phosphorus*	P	2,8,5
6	Carbon*	C	2,4	16	Sulphur*	S	2,8,6
7	Nitrogen*	N	2,5	17	Chlorine*	Cl	2,8,7
8	Oxygen*	O	2,6	18	Argon*	Ar	2,8,8
9	Fluorine*	F	2,7	19	Potassium#	K	***2,8,8,1***
10	Neon*	Ne	2,8	20	Calcium#	Ca	***2,8,8,2***

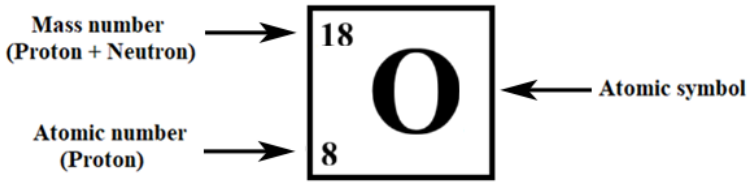
#: Metal **^: Semi-metal** ***: Non-metal**

Atomic number: Number of protons

Mass number: Number of protons + Number of neutrons

Number of protons = Number of electrons (in a neutral atom)

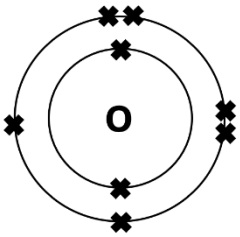
Full Atomic Symbol



Electronic Diagram

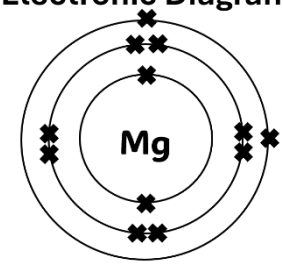
O = 8 = 2,6

Electronic Diagram



Mg = 12 = 2,8,2

Electronic Diagram



Structure Questions (Part2)

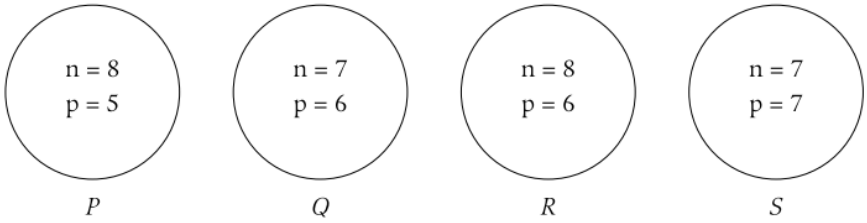
Q1. (a) Complete the following table:

Atomic number	Mass number	No. of proton	No. of neutron	No. of electron	Name of element	Full atomic symbol
1	2					
2			2			
3	6					
4			5			
5	11					
6			8			
7	15					
8			9			
9	19					
10			11			
11	23					
12			13			
13	27					
14			15			
15	31					
16			18			
17	37					
18			22			
19	41					
20			24			

(20 marks)

(b) Which FOUR elements are most abundant in the Earth crust? Arrange them in decreasing order of abundance

Q2: The diagram below shows the nuclei of four atoms P, Q, R and S.



n = number of neutrons;
p = number of protons

(a) State and explain which two atoms belong to the same element.

(2 marks)

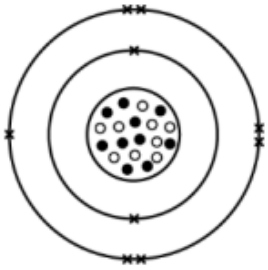
(b) What is the element in (a)?

(1 mark)

(c) Write the full atomic symbol for each of these atoms.

(4 marks)

Q3: The diagram below represents the structure of an atom of element X. The crosses, black dots and white dots represent three different kinds of subatomic particles.



(a) (i) What type of subatomic particle does the black dot represent?

(1 mark)

(ii) State the charge of the black dot.

(1 mark)

(b) Determine the atomic number and the mass number of X.

(2 marks)

(c) Hence, write the full atomic symbol of X.

(1 mark)

Q4: Draw electron diagrams of atoms of the following elements:

(i) Fluorine

(1 mark)

(ii) Calcium

(1 mark)

(iii) Neon

(1 mark)

(iv) Phosphorus

(1 mark)

Q5: Look at the table. It shows some information about the particles which make up atoms.

Particle	Charge	Relative Mass
Proton		1
Electron		0.005
Neutron	Neutral	

(a) Complete the table.

(3 marks)

(b) Look at the symbol for magnesium.

$$^{25}_{12}\mathrm{Mg}$$

The mass number of magnesium is 25.

(i) What is meant by 'mass number'?

(1 mark)

(ii) Write down the electronic arrangement of a magnesium atom.

(1 mark)

(iii) A magnesium atom is electrically neutral. Explain why.

(1 mark)

Ans Key for Part 1:

Q1: No. of positively charged protons = no. of negatively charged electrons

Q2: (i) Electron

(ii) Neutron

(iii) Proton

Q3: (a) No. This is because in an atom, the number of protons (positively charged) is equal to the number of electrons (negatively charged).

(b) Electrons

(c) Protons and neutrons

(d) In an atom, protons and neutrons are concentrated in the nucleus.

Besides, the masses of protons and neutrons are relatively much larger than that of electrons.

Ans Key for Part 2:

Q1a: 20

1	2	***1***	***1***	***1***	***Hydrogen***	^{2}H
2	***4***	***2***	2	***2***	***Helium***	^{4}He
3	6	***3***	***3***	***3***	***Lithium***	^{6}Li
4	***9***	***4***	5	***4***	***Beryllium***	^{9}Be
5	11	***5***	***6***	***5***	***Boron***	^{11}B
6	***14***	***6***	8	***6***	***Carbon***	^{14}C
7	15	***7***	***8***	***7***	***Nitrogen***	^{15}N
8	***17***	***8***	9	***8***	***Oxygen***	^{17}O
9	19	***9***	***10***	***9***	***Fluorine***	^{19}F
10	***21***	***10***	11	***10***	***Neon***	^{21}Ne
11	23	***11***	***12***	***11***	***Sodium***	^{23}Na
12	***25***	***12***	13	***12***	***Magnesium***	^{25}Mg
13	27	***13***	***14***	***13***	***Aluminium***	^{27}Al
14	***29***	***14***	15	***14***	***Silicon***	^{29}Si
15	31	***15***	***16***	***15***	***Phosphorus***	^{31}P
16	***34***	***16***	18	***16***	***Sulphur***	^{34}S
17	37	***17***	***20***	***17***	***Chlorine***	^{37}Cl
18	***40***	***18***	22	***18***	***Argon***	^{40}Ar
19	41	***19***	***22***	***19***	***Potassium***	^{41}K
20	***44***	***20***	24	***20***	***Calcium***	^{44}Ca

Q1b: Oxygen, silicon aluminium, iron

Q2: (a) Q and R

They have the same number of protons.

(b) Carbon

(c) For atom P: ${}^{13}_{5}P$

For atom Q: ${}^{13}_{6}Q$

For atom R: ${}^{14}_{6}R$

For atom S: ${}^{14}_{7}S$

Q3: (a) (i) Neutron

(ii) 0

(b) Atomic number = 9, Mass number = 19

(c) ${}^{19}_{9}F$

Q4: (i) (ii) (iii) (iv)

Q5: (a)

Particle	Charge	Relative Mass
Proton	Positive	1
Electron	Negative	0.005
Neutron	Neutral	1

(b) (i) Mass number of an atom is the sum of the numbers of protons and neutrons in the atom.

(ii) 2, 8, 2

(iii) It has the same number of protons and electrons.

PERIODIC TABLE 週期表

GROUP 族

atomic number 原子序

relative atomic mass 相對原子質量

I	II											III	IV	V	VI	VII	0
				1 **H** 1.0													2 **He** 4.0
3 **Li** 6.9	4 **Be** 9.0											5 **B** 10.8	6 **C** 12.0	7 **N** 14.0	8 **O** 16.0	9 **F** 19.0	10 **Ne** 20.2
11 **Na** 23.0	12 **Mg** 24.3											13 **Al** 27.0	14 **Si** 28.1	15 **P** 31.0	16 **S** 32.1	17 **Cl** 35.5	18 **Ar** 40.0
19 **K** 39.1	20 **Ca** 40.1	21 **Sc** 45.0	22 **Ti** 47.9	23 **V** 50.9	24 **Cr** 52.0	25 **Mn** 54.9	26 **Fe** 55.8	27 **Co** 58.9	28 **Ni** 58.7	29 **Cu** 63.5	30 **Zn** 65.4	31 **Ga** 69.7	32 **Ge** 72.6	33 **As** 74.9	34 **Se** 79.0	35 **Br** 79.9	36 **Kr** 83.8
37 **Rb** 85.5	38 **Sr** 87.6	39 **Y** 88.9	40 **Zr** 91.2	41 **Nb** 92.9	42 **Mo** 95.9	43 **Tc** (98)	44 **Ru** 101.1	45 **Rh** 102.9	46 **Pd** 106.4	47 **Ag** 107.9	48 **Cd** 112.4	49 **In** 114.8	50 **Sn** 118.7	51 **Sb** 121.8	52 **Te** 127.6	53 **I** 126.9	54 **Xe** 131.3
55 **Cs** 132.9	56 **Ba** 137.3	57 * **La** 138.9	72 **Hf** 178.5	73 **Ta** 180.9	74 **W** 183.9	75 **Re** 186.2	76 **Os** 190.2	77 **Ir** 192.2	78 **Pt** 195.1	79 **Au** 197.0	80 **Hg** 200.6	81 **Tl** 204.4	82 **Pb** 207.2	83 **Bi** 209.0	84 **Po** (209)	85 **At** (210)	86 **Rn** (222)
87 **Fr** (223)	88 **Ra** (226)	89 ** **Ac** (227)	104 **Rf** (261)	105 **Db** (262)													

*	58 **Ce** 140.1	59 **Pr** 140.9	60 **Nd** 144.2	61 **Pm** (145)	62 **Sm** 150.4	63 **Eu** 152.0	64 **Gd** 157.3	65 **Tb** 158.9	66 **Dy** 162.5	67 **Ho** 164.9	68 **Er** 167.3	69 **Tm** 168.9	70 **Yb** 173.0	71 **Lu** 175.0
**	90 **Th** 232.0	91 **Pa** (231)	92 **U** 238.0	93 **Np** (237)	94 **Pu** (244)	95 **Am** (243)	96 **Cm** (247)	97 **Bk** (247)	98 **Cf** (251)	99 **Es** (252)	100 **Fm** (257)	101 **Md** (258)	102 **No** (259)	103 **Lr** (260)